Chengpo Yan

jesse-yan.github.io | cyan46@wisc.edu

Education

University of Wisconsin–Madison Madison, WI
Ph.D. in Computer Sciences, Expected May 2028 *Sep. 2023 – Present*

University of Wisconsin–Madison Madison, WI
B.S. in Computer Sciences & Data Science *Sep. 2019 – Dec. 2022*

Research Interests

Machine learning systems, with a focus on efficient LLM inference and serving on heterogeneous accelerators.

Publications

[1] Bangya Liu, **Chengpo Yan**, Chenghao Jiang, Suman Banerjee, and Akarsh Prabhakara. "Privacy-Aware Sharing of Raw Spatial Sensor Data for Cooperative Perception." In *Proceedings of the 27th International Workshop on Mobile Computing Systems and Applications (HotMobile)*, 2026.

[2] William Dong, Jason Lian, **Chengpo Yan**, Yiran Zhong, Sumanth Karnati, Qilin Guo, Lianyi Chen, and Dane Morgan. "Deep-Learning-Based Segmentation of Keyhole in In-Situ X-ray Imaging of Laser Powder Bed Fusion." *Materials*, 2024.

[3] Saurabh Agarwal, **Chengpo Yan**, Ziyi Zhang, and Shivaram Venkataraman. "Bagpipe: Accelerating Deep Recommendation Model Training." In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.

Preprints

[1] Sihyeon Lee, Hojeong Lee, Sungwon Woo, **Chengpo Yan**, Suman Banerjee, and Seyeon Kim. "NPUsper: Eliminating Redundant Computation for Real-Time Whisper on Mobile NPUs." *arXiv:2607.01108*, 2026.

Research Experience

Graduate Researcher with Prof. Suman Banerjee Madison, WI
Machine Learning Systems and Wireless Sensing *Sep. 2024 – Present*

- Designing speculative decoding methods and lightweight drafter architectures for efficient LLM inference.
- Building on-device LLM inference systems across heterogeneous mobile GPUs and NPUs.
- Developing mmWave sensing systems for contactless physiological monitoring in clinical settings.

Undergraduate Researcher with Prof. Shivaram Venkataraman Madison, WI
Bagpipe: Accelerating Deep Recommendation Model Training *May 2022 – Dec. 2022*

- Developed distributed embedding caching and writeback mechanisms for deep recommendation model training, reducing parameter-server I/O by 15%.
- Overlapped NCCL cache synchronization with model computation, reducing training time by 10%; the overall Bagpipe system achieved up to 5.6× speedup over state-of-the-art baselines.

Researcher with Prof. Suman Banerjee Madison, WI
Wi-Fi and Bluetooth Indoor Localization *Jul. 2022 – Dec. 2023*

- Developed RSSI/CSI-based indoor localization methods, achieving a median error of 2 m.

Undergraduate Researcher with Prof. Dane Morgan Madison, WI
Machine Learning for In-Situ X-ray Imaging *Mar. 2022 – Dec. 2022*

- Developed an image segmentation and measurement pipeline for in-situ X-ray imaging of laser powder bed fusion.

Professional Experience

University of Wisconsin–Madison Madison, WI
Teaching Assistant *Sep. 2023 – Present*

- Teaching assistant for CS 354: Machine Organization and Programming (Spring 2024–Summer 2026) and CS 540: Introduction to Artificial Intelligence (Fall 2023).

Skyworth Shenzhen, China
Software Engineer Intern *May 2021 – Aug. 2021*

- Developed and optimized Swaiot, a cross-platform AIoT application for Skyworth devices, reducing load time by 9% and response latency by 11%.

Technical Skills

Languages: Python, C/C++, Java
ML & Systems: PyTorch, CUDA, NCCL, SGLang, vLLM, Qualcomm QNN, LiteRT, Linux, Docker

Awards

2nd Place, STAI-X Challenge — Award B: Autonomous Data-Analysis Agent, 2026
Meritorious Winner, Mathematical Contest in Modeling (MCM), 2018